



Hybrid supercapacitors from porous boron-doped diamond with water-soluble redox electrolyte

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ABSTRACT

Hybrid supercapacitors (HSCs) can simultaneously achieve battery-level energy as well as capacitor-level power and cycle life in a single device. To construct such HSCs, porous boron-doped diamond (BDD) films fabricated by a template-free method via regrowth of diamond nanoparticles were employed as the electrodes and water-soluble redox couples ($\text{Fe}(\text{CN})_6^{3-/4-}$) as the electrolyte. For the three-electrode system, the porous BDD electrode delivers a capacitance of 13.08 mF cm^{-2} in $1.0 \text{ M Na}_2\text{SO}_4$, and a capacity of 78.23 mAh g^{-1} in $0.05 \text{ M Fe}(\text{CN})_6^{3-/4-} + 1.0 \text{ M Na}_2\text{SO}_4$ when a scan rate of 10 mV s^{-1} is applied. For the symmetric two-electrode system, the porous BDD electrode exhibits a capacitance of 13.04 mF cm^{-2} in $1.0 \text{ M Na}_2\text{SO}_4$ and a capacity of 34.74 mAh g^{-1} in $0.05 \text{ M Fe}(\text{CN})_6^{3-/4-} + 1.0 \text{ M Na}_2\text{SO}_4$ at the same scan rate. The capacitance (capacity) remains about 92% of its initial value after 5000 cyclic voltammetry cycles in all conditions. The specific power densities (normalized by the mass of electrode active material only) are 0.17 and 1.05 kW kg^{-1} , with their specific energy densities of 5.51 and 38.10 Wh kg^{-1} , respectively. Thus, the porous BDD films, together with the water-soluble redox electrolyte, have profound applications in high-performance hybrid supercapacitors construction for future power devices.

1. Introduction

The energy crisis threatens human beings with the exhaustion of fossil fuels as well as greenhouse-gas emissions [1,2]. To address these concerns, sustainable and renewable energy sources, such as solar, wind, and hydro energy have been well developed [3–5]. However, the storage of electricity obtained from the above-mentioned energy supplies by electrochemical energy-storage devices is still a great challenge due to the large fluctuations when electricity generations [6,7]. Among the various electrochemical energy storage devices, batteries and supercapacitors serve the mainstream via different charge-storage mechanisms [8]. Batteries, especially lithium-ion batteries, can deliver a high energy density as high as 200 Wh kg^{-1} , which are widely used in consumer electronics [9]. However, the heat generation gives rise to serious safety issues, especially at high power conditions [10,11]. What's more, batteries are also confronted with a low power density as well as short service life [12]. On the other hand, supercapacitors can safely offer higher power density, faster-charging process, and longer capacitance retention, which can be a complement or even replacement

of batteries in some applications if the weakness of low energy density could be well solved [8,13,14]. Thus, a hybrid supercapacitor both with the battery-level energy density, as well as the supercapacitor-level power density and long-term stability is an urgent demand [15].

Extensive research efforts are underway to address the challenge of relatively low energy density that the supercapacitors are facing [8,16–18]. According to the equation of energy density calculation, it can be enhanced through both capacitance increase as well as potential window enlargement [19]. Thus, three-dimensional (3D) porous carbon with a large specific area [20], pseudocapacitive materials with a high theoretical capacitance [17] as well as asymmetric configuration devices with a wide potential window [21,22] are employed and great achievements have been made. Thus, the first issue in the fabrication of hybrid supercapacitors with high electrochemical performance is the choice of suitable capacitor electrodes. Among various electrode materials, boron-doped diamond (BDD) with a porous nanostructure is one of the ideal candidates [23–25]. The BDD electrode has much wider potential windows in various electrolytes than most of the electrode materials, which can greatly enhance the power density [23].

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Meanwhile, the BDD electrode fabricated via top-down etching methods [26–28] or bottom-up procedure with [29,30] (or without [19,31]) a template achieves a large specific area, thus, a high capacitance can be attained. Moreover, the unique sp^3 hybrid structure endows the BDD with super inert characteristic and long-term stability even in extremely harsh conditions [32].

For construction hybrid supercapacitors, another critical issue to be considered is the selection of electrolytes. The introduction of organic electrolytes, ionic liquids as well as solid-state polymer electrolytes can broaden the stable potential window and enhance the electrochemical performance [33–35]. However, these electrolytes also suffer from some fatal flaws, such as the high cost, highly dangerous, and environmentally unfriendly. Thus, a low-cost, safe, and environmentally friendly aqueous electrolytes is an efficient solution. Redox species, such as polymers and metal-oxides, which are coated on conductive substrates, are widely used as a pseudocapacitor electrode to enhance the capacitance [21,36]. Among them, the $Fe(CN)_6^{3-/4-}$ redox couples, which can be dissolved in aqueous solution, is of great advantage [19,37,38]. Moreover, the charge storage capacity is easy to be controlled by tuning the amount of the redox species [19].

In this work, we firstly introduce a novel and straightforward template-free method to fabricate a porous BDD film by microwave plasma chemical vapor deposition (MPCVD) via the regrowth of the self-assembly diamond nanoparticles. The characterizations of the surface morphology, specific surface area, pore size distribution, crystallographic and chemical structure of the film by scanning electron microscopy (SEM), N_2 adsorption/desorption isotherm, X-ray diffraction (XRD), Raman spectrum, and X-ray photoelectron spectroscopy (XPS) were followed. To evaluate the electrochemical performance of the as-grown porous BDD electrode, both the three-electrode system and symmetrical two-electrode system were employed. For electrochemical double layer capacitors (EDLC), all the measurements, including cyclic voltammetry (CV), galvanostatic charge/discharge (GCD), and electrochemical impedance spectroscopy (EIS) were conducted in 1.0 M Na_2SO_4 solution. While for hybrid supercapacitors (HSC) with high capacity, the redox couples of 0.05 M $Fe(CN)_6^{3-/4-}$ were specially dissolved in 1.0 M Na_2SO_4 solution, and the electrochemical performance was studied using the above-mentioned techniques. Moreover, the stability of the porous BDD electrodes was evaluated by CV during a 5000-time cycle both in 1.0 M Na_2SO_4 and 0.05 M $Fe(CN)_6^{3-/4-}$ + 1.0 M Na_2SO_4 , in single and device configurations. At last, the specific energy and power densities (normalized by the mass of deposited porous BDD) of EDLC and HSC were compared.

2. Experimental section

2.1. Electrode material

Porous BDD films were deposited on silicon wafers ($2 \times 2 \text{ cm}^2$) using the MPCVD technique. The as-received detonation diamond nanoparticles (5–10 nm, N140013, Aladdin Industrial Corporation, China) were dispersed in ultrapure water during 60 min ultrasonic treatment to form a semi-uniform colloidal suspension with a concentration of 2 g L^{-1} for diamond synthesis seeding. A flat and continuous water film was formed as the prepared suspension was dropped onto the silicon wafer. After the suspension film was dried naturally at room temperature, a porous layer of diamond nanoparticles was gained by self-assembly. The porous BDD film growth was conducted at $950 \pm 10^\circ\text{C}$, 10 kPa, and 3000 W with a duration of 15 min. The mass loading of the porous BDD was determined by subtracting the weight of the naked silicon substrate from the total weight after seeding and deposition, and a mass loading of 0.6 mg cm^{-2} was gained. A mixture of 4 standard cubic centimeters per minute (sccm) CH_4 and 200 sccm H_2 was used as the main precursor. Boron was doped into the film by passing an additional H_2 (20 sccm) through a bubble containing tributyl borate (Aladdin Industrial Corporation, China) as the boron source.

2.2. Characterization

The surface morphologies of the porous BDD films were investigated by field emission scanning electron microscopy (FE-SEM, ZEISS Merlin VP Compact). A GIXRD (M189XHF-SRA, Mac Science), equipped with Cu $K\alpha$ radiation ($\lambda = 0.15406 \text{ nm}$) was employed to analyze the crystallographic structure of the films. The Raman spectrum of the film was recorded on a Raman spectrometer (HR-800, Horiba, Jobin Yvon, Japan) at room temperature with a 532 nm laser excitation. An X-ray photoelectron spectroscopy (XPS) was used to characterize the elemental composition of the film. Specific surface area and the pore distribution of the porous BDD film were estimated by the Brunauer-Emmett-Teller (BET) method and the density functional theory (DFT) program using an AUTOSORB-1 surface area analyzer (Quantachrome Instrument Corporation).

2.3. Electrochemical measurements

The CV, GCD, and EIS behaviors of the porous BDD electrode were conducted on a CS2350 electrochemical workstation (Wuhan CorrTest Instruments Corporation, China) with different potential windows, scan rates, and current densities. A standard three-electrode cell, where a porous BDD film (immersed geometrical area: 2 cm^2 , active material mass loading: 0.6 mg cm^{-2}) acts as the working electrode, an Ag/AgCl (in 3.0 M KCl) electrode as the reference electrode, high-purity platinum sheet as the counter electrode, and 1.0 M Na_2SO_4 (or 0.05 M $K_3Fe(CN)_6/K_4Fe(CN)_6$ + 1.0 M Na_2SO_4) (80 ml) as the electrolyte, was utilized to test the single BDD electrode electrochemical performance. For the full-cell performance test, a two-electrode symmetric supercapacitor device was constructed (Fig. S1), with two porous BDD films (immersed geometrical area of one single porous BDD electrode: 2 cm^2 , active material mass loading: 0.6 mg cm^{-2}) served as the capacitor electrodes, a 50 μm thick Nafion film (Alfa Aesar) as the separator, and 1.0 M Na_2SO_4 (or 0.05 M $K_3Fe(CN)_6/K_4Fe(CN)_6$ + 1.0 M Na_2SO_4) (2.6 ml) as the electrolyte. The EIS was measured in the frequency ranging from 100 kHz to 0.1 Hz at an open circuit potential with an amplitude of 5 mV. Stability tests were performed using CV during 5000 cycles.

Considering that the CV and GCD curves of porous BDD in Na_2SO_4 electrolyte show EDLC behavior, the charge storage performance of porous BDD in Na_2SO_4 electrolyte was evaluated using specific capacitance (C_s , mF cm^{-2}) [39]. When redox couples were added, the porous BDD electrodes exhibit obvious Faradic peaks and charge/discharge plateaus [40]. Thus, the charge storage performance was evaluated by specific capacity (Q_s , mAh g^{-1}) [41]. They can be calculated from the CV and GCD curves by using the following equations, respectively [42].

$$C_{s,CV} = \frac{1}{2} \frac{\int I(V) dV}{\Delta V \times v \times A} \quad (1)$$

$$C_{s,GCD} = \frac{I \times \Delta t}{\Delta V \times A} \quad (2)$$

$$Q_{s,CV} = \frac{1}{2} \frac{\int I(V) dV}{3.6 \times v \times m} = \frac{1}{7.2} \frac{\int I(V) dV}{v \times m} \quad (3)$$

$$Q_{s,GCD} = \frac{1}{3.6} \frac{\int I(V) dt}{m \times \frac{\Delta V}{2}} = \frac{1}{1.8} \frac{\int I(V) dt}{m \times \Delta V} \quad (4)$$

where $C_{s,CV}$, $C_{s,GCD}$, $Q_{s,CV}$, $Q_{s,GCD}$, $I(V)$, ΔV , v , A , I , Δt , m , and $I(V)$ are the specific capacitance obtained from CVs, GCDs, the specific capacity obtained from CVs, GCDs, operating current, potential window, scan rate, geometric area, discharge current, discharge time, mass and operating potential of a single porous BDD electrode, respectively. When the above-mentioned equations are used to evaluate the assembled symmetrical device's performance, the geometric area and mass are

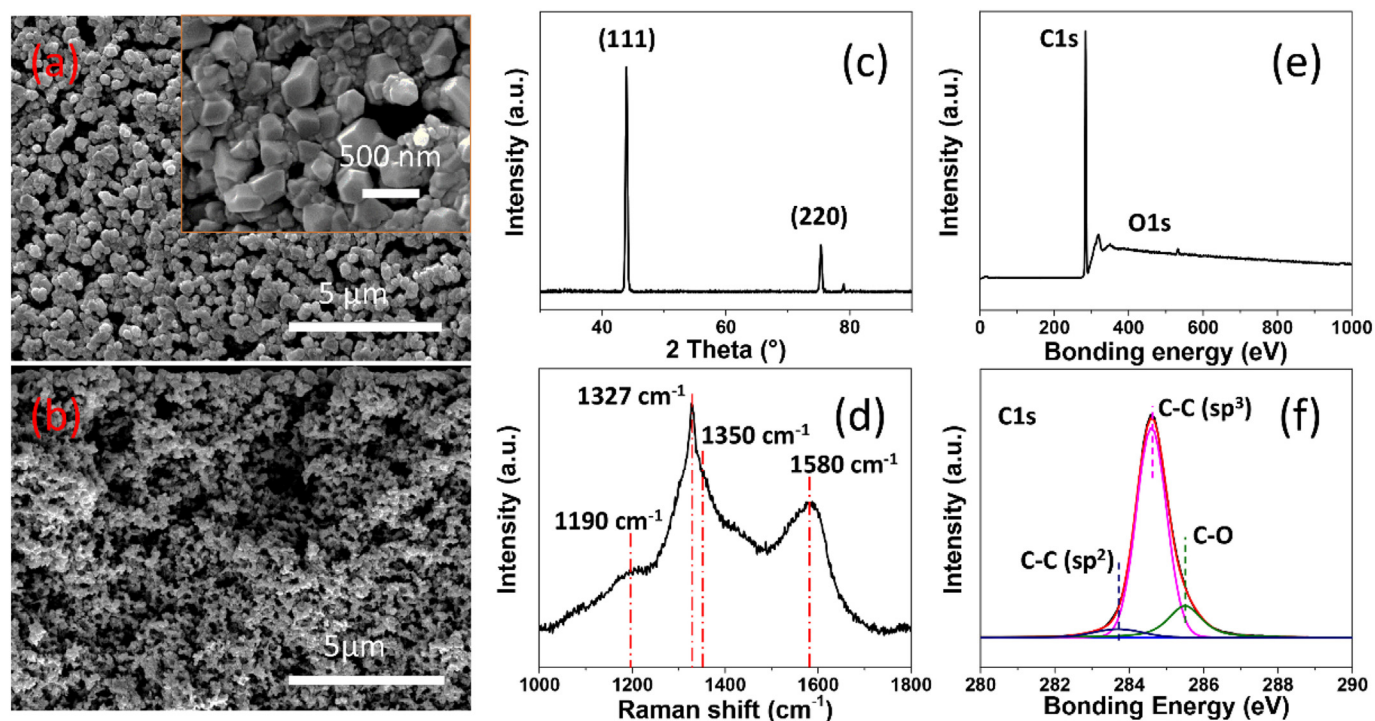


Fig. 1. (a) Top, (b) cross-section SEM images, (c) XRD pattern, (d) Raman spectrum, (e) XPS survey spectrum, and (f) high-resolution XPS of C1s for porous BDD film, the inset image of (a) shows the higher magnification view of the film.

doubled considering the positive and negative electrodes. For the assembled symmetrical device, the specific energy and power densities were calculated via the following equations, respectively [43].

$$E = \frac{I \int V(t) dt}{3.6 \times M} \quad (5)$$

$$P = \frac{E}{\Delta t} \quad (6)$$

where E , P , I , $V(t)$, M , and Δt are the specific energy density, specific power density, discharge current, operating potential, the total mass of the positive and negative porous BDD electrodes (the mass of separator and electrolyte are excluded), and discharge time, respectively.

3. Results and discussion

3.1. Morphology, orientation and chemical composition of porous BDD film

A porous BDD film was employed as the capacitor electrode. Fig. 1a and b shows the typical SEM images of the used porous BDD film during a 15-minute growth. The film presents a three-dimensional (3D) and interconnected porous nanostructure with the grain sizes varying from tens to hundreds of nanometers. Abundant nano-sized pores with nonuniform size ranging from several to hundred nanometers and different shapes are randomly distributing between the diamond grains. The 3D and interconnected porous nanostructure, combining with the multilevel pore distribution, can effectively increase the specific area, benefit the ion adsorption/desorption, shorten the ion transportation path, decreases the impedance, and enhance the electrochemical performance when used as electrode materials [44]. As determined by N_2 adsorption/desorption isotherm (Fig. S2), the porous BDD film shows a wide range of pore size distribution and a specific area of about $282 \text{ m}^2 \text{ g}^{-1}$ with a type IV hysteresis loop, corresponding to the characteristic of mesoporous materials [45]. Fig. 1c shows the X-ray diffraction (XRD) pattern of the as-prepared porous BDD film, both the sharp peaks at the 2θ of 43.9° and 75.3° can be well indexed to the (111) and (220) diamond planes and are in well agreement with those for diamond (PDF

No. 06-0675), indicating the dominant presence of diamond phase. To further confirm the diamond composition, Raman and X-ray photoelectron spectroscopy (XPS) measurements were carried out and the results are shown in Fig. 1d–f. The Raman spectrum consisting of peaks at $\sim 1190 \text{ cm}^{-1}$, as well as the downshift and broadening of the intrinsic sp^3 diamond peak at $\sim 1327 \text{ cm}^{-1}$, indicates the porous BDD film is dominant by diamond phase and middle-content boron-doped (in the range of $10^{18-19}/\text{cm}^{-3}$) [46,47] with a room temperature resistivity of $18 \Omega \text{ cm}$. The porous BDD film also shows a shoulder at around 1350 cm^{-1} and a broadening band between 1500 and 1650 cm^{-1} , which are attributed to the disordered and ordered sp^2 carbon [48]. The films that grown within a short duration are nanocrystalline in nature and plenty of grain boundaries in high sp^2 content are existing. Moreover, the non-diamond phases are hindered from an etching by the reactive hydrogen radicals by the complex 3D and interconnected nanostructure [48]. The survey scan XPS spectrum shows peaks at about 284.8 and 533 eV are attributed to the photoelectrons excited from the C1s and O1s core levels, respectively [49]. The presence of oxygen may be due to the samples are gradually oxidized as they are exposed to air [23], as well as the boron source is oxygen-involving. The C1s spectrum is deconvoluted by the Gaussian-Lorentzian method and the fitted result is shown in Fig. 1f. The peaks located at 283.7 , 284.6 , and 285.5 eV are attributed to the C–C sp^2 , C–C sp^3 and C–O hybridized carbon atoms, respectively [50]. What's more, the sp^3 fraction is more than 85%, further confirming the porous BDD film is diamond phase dominant. It is worth noting that the films are oxygen-terminated, differs from most of the hydrogen-terminated diamond films via the MPCVD method, and this feature endows the film with hydrophilic, which will enhance the specific area, thus, a higher capacitance can be gained [51].

3.2. Electrochemical performance

The EDLC capacitive behavior of a porous BDD electrode was firstly investigated in $1.0 \text{ M Na}_2\text{SO}_4$ within the potential range from 0 to 0.8 V (vs. Ag/AgCl) by CVs in a three-electrode system. The CVs of porous

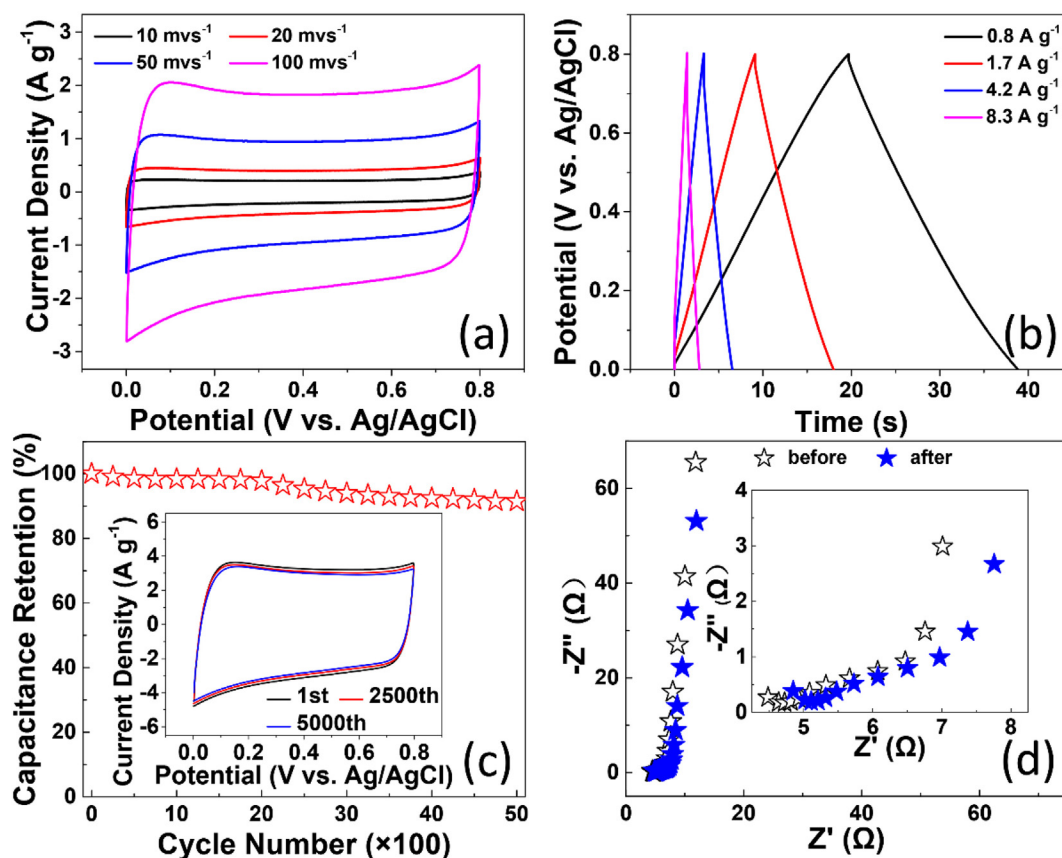


Fig. 2. (a) CVs, (b) GCDs for porous BDD electrode recorded in 1.0 M Na₂SO₄ at the scan rates of 10, 20, 50, 100 mV s⁻¹, and at the current densities of 0.8, 1.7, 4.2, and 8.3 A g⁻¹, respectively. (c) The stability test during 5000-cycle CV scanning, the inset shows the 1st, 2500th, and 5000th cycle. (d) Nyquist plots for the porous BDD electrode before and after 5000-cycle CV scanning, the inset shows the detail in the high-frequency area of porous BDD electrodes before and after the stability test.

BDD electrode with different scan rates are recorded in Fig. 2a, which shows a rectangular shape. Moreover, the current densities increase with the scan rate and the current density is linear dependent on the scan rate, indicating the typical feature of an ideal EDLC [14]. The estimated specific capacitances are 13.08, 12.53, 11.64, and 10.98 mF cm⁻² (21.80, 20.89, 19.40, and 18.29 F g⁻¹) at the scan rates of 10, 20, 50, and 100 mV s⁻¹, respectively, which are higher than most of the reported diamond nanostructures via top-down etching methods and bottom-up growth process [19,30,52]. The capacitance of the porous BDD electrode is further evaluated by GCD measurement as shown in Fig. 2b. All GCDs show symmetric and linear shape, indicating the porous BDD electrode with high reversibility and coulombic efficiency. The estimated specific capacitances are 11.98, 11.09, 10.08, and 8.66 mF cm⁻² (19.97, 18.49, 16.80, and 14.43 F g⁻¹) at the current densities of 0.5, 1.0, 2.5, and 5.0 mA cm⁻² (0.8, 1.7, 4.2, and 8.3 A g⁻¹), respectively. The capacitance values obtained from GCD are consistent with the values obtained from CV scanning. Apart from the high capacitance achieved, the porous BDD electrode also features outstanding cycling stability as shown in Fig. 2c, which is conducted in 1.0 M Na₂SO₄ via a CV technique at a scan rate of 200 mV s⁻¹ during 5000 cycles. The CVs exhibit a stable and rectangular shape with a negligible area shrinkage and inconspicuous deformation. After 5000 cycles, the electrode remains 91.4% of the initial capacitance. To explain the superior electrochemical properties of the porous BDD electrode, the impedance is employed as a powerful tool. The Nyquist impedance plots (Fig. 2d), which are composed of a little arc in high frequencies and a straight line in low frequencies [53], show negligible differences both before and after stability test, confirming the excellent stability of the porous BDD electrode. The equivalent series resistance

(ESR), which sums up the resistance of the electrolyte, the electrode, as well as the contact resistance between the electrode and the current collector, is less than 2.5 Ω cm⁻², indicating the high conductivity of BDD and the well electric contact between BDD and the silicon wafer.

To achieve battery-level energy density as well as supercapacitor-level power density, the BDD HSCs were fabricated via adding the water-soluble redox species (0.05 M Fe(CN)₆^{3-/4-}) into 1.0 M Na₂SO₄ to further enhance the electrochemical performances. The CVs within the potential window ranging from -0.5 to 0.9 V (vs. Ag/AgCl) in Fig. 3a shows a pair of remarkable peaks at all scan rates, corresponding to the Faradic reaction of [Fe(CN)₆]³⁻ + e⁻ ↔ [Fe(CN)₆]⁴⁻ [38]. As can be seen, the oxidation peak currents are almost identical to the reduction peak currents, indicating the good reversibility of the BDD HSCs [19]. However, the redox peak potential separation (ΔE_p) of Fe(CN)₆^{3-/4-} redox couple increases from 252 to 553 mV as the scan rate varying from 10 to 100 mV s⁻¹, which is higher than some of the reported diamond nanostructures [54,55]. The variation may be due to the porous BDD film is middle-level boron-doped, which slows down the electron transfer rates of the redox reaction and alters the interactions between redox probes and the porous BDD electrode [51]. The estimated capacities in water-soluble redox electrolyte are 78.23, 63.59, 40.91, and 29.26 mAh g⁻¹ at the scan rates of 10, 20, 50, and 100 mV s⁻¹, respectively, suggesting the high electrochemical performance of HSCs. The related GCD curves are also presented in Fig. 3b. All of these recorded GCD curves show nonlinear shape with redox reactions plateaus of Fe(CN)₆^{3-/4-}, which vary with the applied current densities. Moreover, the times for the charging processes are almost identical to the discharging at all current densities, demonstrating the HSCs of good reversibility. The capacities are 107.79, 56.82, 37.47,

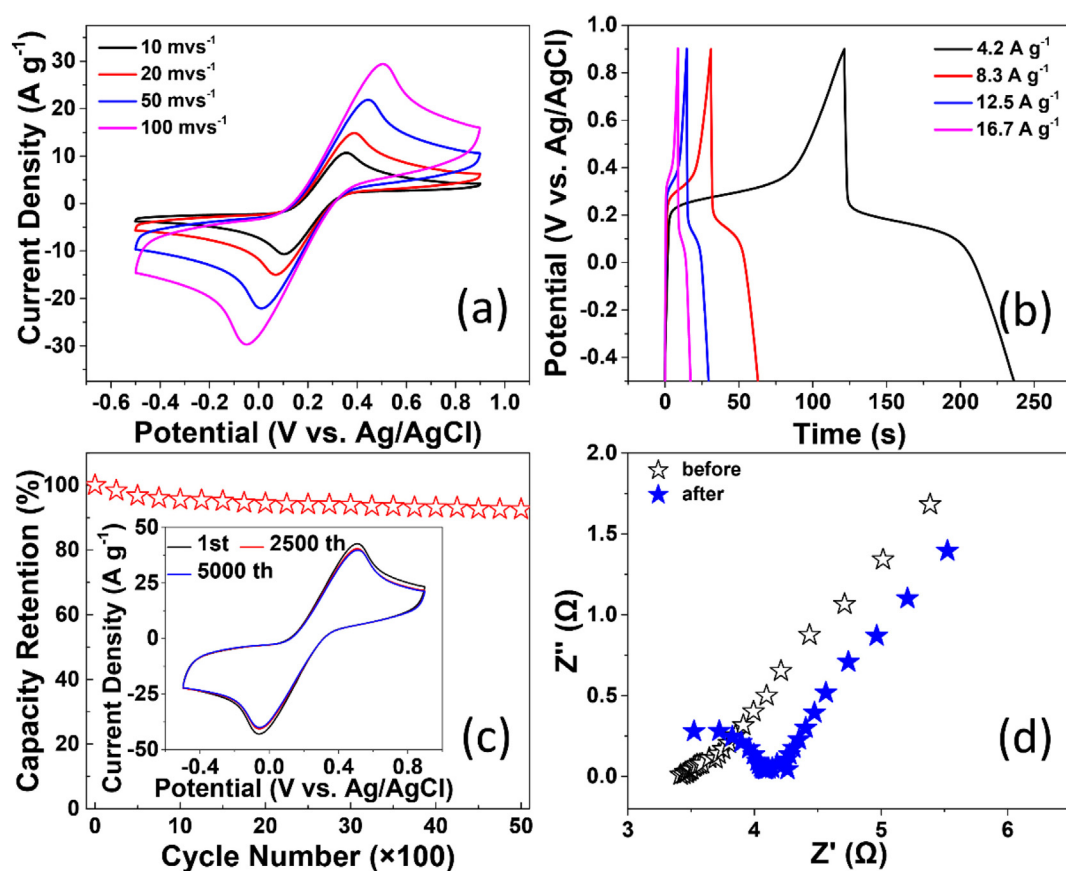


Fig. 3. (a) CVs, (b) GCDs for porous BDD electrode recorded in 0.05 M $\text{Fe}(\text{CN})_6^{3-/4-}$ + 1.0 M Na_2SO_4 at the scan rates of 10, 20, 50, 100 mV s^{-1} and at the current densities of 4.2, 8.3, 12.5, and 16.7 A g^{-1} , respectively. (c) The stability test during 5000-cycle CV scanning, the inset shows the 1st, 2500th, and 5000th cycle. (d) Nyquist plots for the porous BDD electrode before and after 5000-cycle CV scanning.

and 27.33 mAh g^{-1} at the current densities of 4.2, 8.3, 12.5, and 16.7 A g^{-1} , respectively. Notably, most of the energy stored in the HSCs comes from the redox reaction of the $\text{Fe}(\text{CN})_6^{3-/4-}$. Meanwhile, the porous BDD endows the redox couples with more reaction active sites and enables the faster electron transfer between $\text{Fe}(\text{CN})_6^{3-/4-}$ and porous BDD electrode. Moreover, the rich pores from the interconnected network act as micro ion reservoirs, and shorten the distance of the ion diffusion. Thus, the synergistic effect of the high capacity of $\text{Fe}(\text{CN})_6^{3-/4-}$ and the high specific area, multi-scaled of the porous BDD with interconnected network endows the HSCs with high electrochemical performance.

Besides the high specific area as well as the multi-scaled and interconnected porous structure, the inert and stable sp^3 bonding nature of the diamond also endows the HSCs with high cycling stability. Thus, the cycling stability measurement was conducted for further evaluation of the porous BDD HSCs and the results are recorded in Fig. 3c. The CVs were tested in 0.05 M $\text{Fe}(\text{CN})_6^{3-/4-}$ + 1.0 M Na_2SO_4 with 5000 cycles at a scan rate of 200 mV s^{-1} . The capacity loss is about 5% during the first 1000 cycles. However, the capacity almost remains unchanged during the next 4000 cycles, and keeps 92.6% of its initial value, illustrating the excellent cycling stability of the HSCs using the high inert diamond as the electrode as well as the well-reproducible and high-capacity redox couples of $\text{Fe}(\text{CN})_6^{3-/4-}$ as the electrolyte. Impedance was measured and recorded in Fig. 3d when $\text{Fe}(\text{CN})_6^{3-/4-}$ was added. It is notable that the whole resistance obtained from HSCs becomes much smaller than that from EDLCs, and the straight-line part of the HSCs is about 45° tilted while in EDLCs it appears perpendicular to the real axis, demonstrating the fast ion/charge transfer process and battery-like characteristics of redox couples. Moreover, the ESR becomes a little bigger after 5000 CV cycles than before, which may result in the

7.4% capacitance loss during stability test.

For practical applications of the porous BDD electrode, a two-electrode symmetrical supercapacitor device was assembled, in which the as-synthesized porous BDD films were used as capacitor electrodes, a Nafion film as the separator, as well as Na_2SO_4 solution and $\text{Fe}(\text{CN})_6^{3-/4-}$ + Na_2SO_4 as the electrolytes to construct EDLCs and HSCs devices. The potential window for EDLCs was optimized at 2.6 V (vs. Ag/AgCl), while for HSCs, it was set to 2 V (vs. Ag/AgCl) from -1 to 1 V.

The CVs of EDLCs device at various scan rates from 10 to 100 mV s^{-1} are shown in Fig. 4a. The CVs in symmetrical configuration show nearly quasi-rectangular shape at all scan rates, and without any faradic peaks and obvious deformation observed, which is similar to the single electrode test in the three-electrode cell, indicating the good EDLC feature [14]. The estimated capacitances are about 13.04, 10.42, 7.34, and 5.33 mF cm^{-2} (21.74, 17.37, 12.24, and 8.89 F g^{-1}) at the scan rate of 10, 20, 50, and 100 mV s^{-1} , respectively. The porous BDD EDLCs device also behave fast charge/discharge character even at a scan rate as high as 1000 mV s^{-1} (data are not shown), a good shape can be kept and a capacitance of about 3 mF cm^{-2} (5 F g^{-1}) can be retained. The loss of the capacitance at high scan rate is due to some micro-pores are not fully accessible to ions and the not-well hydrophilic nature of the diamond [44]. The related GCDs of EDLCs device were also conducted at the current densities from 0.25 to 2.5 mA cm^{-2} , and the charge/discharge curves were recorded in Fig. 4b. The GCD curves show asymmetrical shape with a lower coulombic efficiency compared with the single electrode test, which takes a longer time for charging and shorter duration for discharging. The phenomenon is more obvious when lower current densities are applied. It may result from the increase of the resistance as shown in Fig. 4d. For the EDLCs device, both the ESR and the charge transfer resistance (R_{ct}) increase. Moreover, the

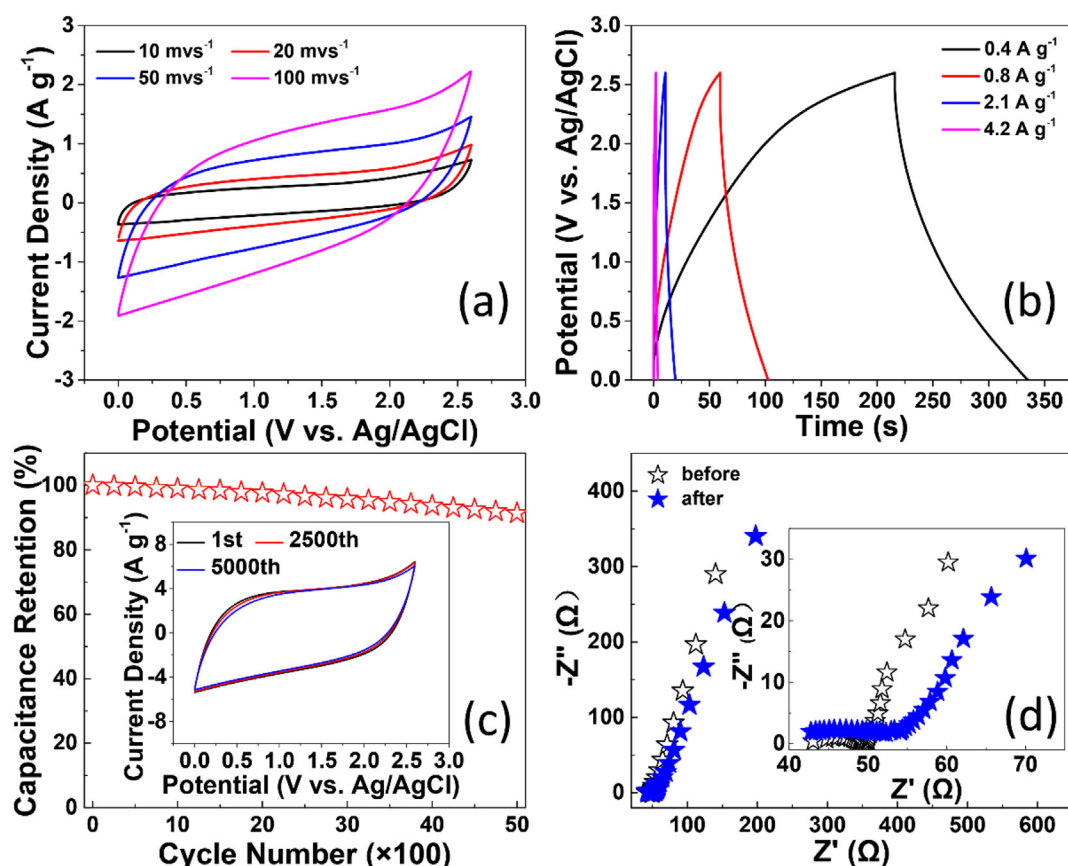


Fig. 4. (a) CVs, (b) GCDs for porous BDD electrode symmetric device recorded in 1.0 M Na_2SO_4 at different scan rates and different current densities. (c) The stability test during 5000-cycle CV scanning, the insets show the 1st, 2500th, and 5000th cycle. (d) Nyquist plot for the porous BDD symmetric device before and after 5000-cycle CV scanning, the inset shows the detail in the high-frequency area of porous BDD electrode symmetric device before and after stability test.

straight-line part tilts with a small angle, instead of almost perpendicular to the real axis. The calculated specific capacitances are about 11.44, 8.27, 4.29, and 1.69 mF cm^{-2} (19.07, 13.79, 7.16 and 2.82 F g^{-1}) at the current densities of 0.25, 0.5, 1.25, and 2.5 mA cm^{-2} (0.4, 0.8, 2.1, and 4.2 A g^{-1}), respectively. Furthermore, the EDLCs device also behaves high stability as the single electrode does, and the results are shown in Fig. 4c. After 5000 CV cycles, the capacitance retention gradually decreases from 100% to 91.8%, and only a negligible area shrinkage is observed, suggesting the excellent stability of the device. The outstanding stability can also be explained by the Nyquist plot as shown in Fig. 4d, which shows a small impedance increase after 5000 CV cycles.

To further evaluate the electrochemical performances of HSCs device, the regular measurements, such as CV, GCD, EIS, and cycle-life test were employed. Fig. 5a shows the CV scanning within the potential window from -1 to 1 V (vs. Ag/AgCl) at different scan rates. The CV curves show a pair of well-shaped oxidation and reduction peaks corresponding to the redox reactions of $\text{Fe}(\text{CN})_6^{3-/4-}$ at all scan rates, in which the anodic peak currents are almost identical to the absolute values of the cathodic peak currents, suggesting the porous BDD HSCs device of good reversibility [38]. Moreover, the ΔE_p increases with the scan rate due to the insufficient dope level and the O-terminated nature of the BDD film [56]. The specific capacities are about 34.74, 29.19, 19.94, and 14.22 mAh g^{-1} at the scan rates of 10, 20, 50, and 100 mV s^{-1} , respectively. The related GCD curves are shown in Fig. 5b, which shows non-linear shapes with charging/discharging plateaus corresponding to the redox reactions of $\text{Fe}(\text{CN})_6^{3-/4-}$ [38]. What's more, the plateaus vary with the current densities in the same trend as HSCs single electrode does in the three-electrode system. The calculated capacities are about 38.10, 20.71, 14.17, and 10.63 mAh g^{-1} at the

current densities of 1.7, 3.3, 5, and 6.7 A g^{-1} , respectively, which is in accord with the values obtained from CV calculation. Apart from the high specific capacities achieved, the HSCs device by the introduction of water-soluble $\text{Fe}(\text{CN})_6^{3-/4-}$ as electrolyte also shows high stability. The stability test was conducted in the form of CVs at the scan rate of 200 mV s^{-1} during 5000 cycles and the related capacity retention vs. cycle numbers is shown in Fig. 5c. The HSCs device remains 91.8% of its initial capacity value, which is comparable to the value of the EDLCs device, EDLCs, and HSCs single electrode. The excellent stability can be confirmed by the Nyquist plots in Fig. 5d before and after 5000 cycles, in which the ERS and R_{ct} show a delicate difference.

As the key criteria to evaluate the performance of the EDLCs and HSCs devices, the specific energy and power densities (normalised by porous BDD mass only) were calculated according to the Eqs. (5) and (6) based on the charging/discharging curves at varied current densities, and the results are shown in Fig. 6 in the form of Ragone plots. The EDLC device delivers a maximum specific energy density of 5.51 Wh kg^{-1} at a specific power density of 167 W kg^{-1} , and remains an energy density of 0.82 Wh kg^{-1} at a power density of 1.67 kW kg^{-1} , demonstrating the high EDLC performance of porous BDD as electrode material. When $\text{Fe}(\text{CN})_6^{3-/4-}$ is added to construct HSCs, the device delivers a maximum specific energy density of 38.10 Wh kg^{-1} with a specific power density of 1.05 kW kg^{-1} and keeps a specific energy density of about 10.63 Wh kg^{-1} at a specific power density of 4.30 kW kg^{-1} . The electrochemical performance of both EDLCs and HSCs were compared with the reported BDD nanostructures, and the results are listed in Table S1. As can be seen, the porous BDD electrode fabricated by our straightforward method exhibits the highest capacitance and specific energy density and excellent cycle stability when used as EDLCs. For HSCs, due to different evaluation methods were

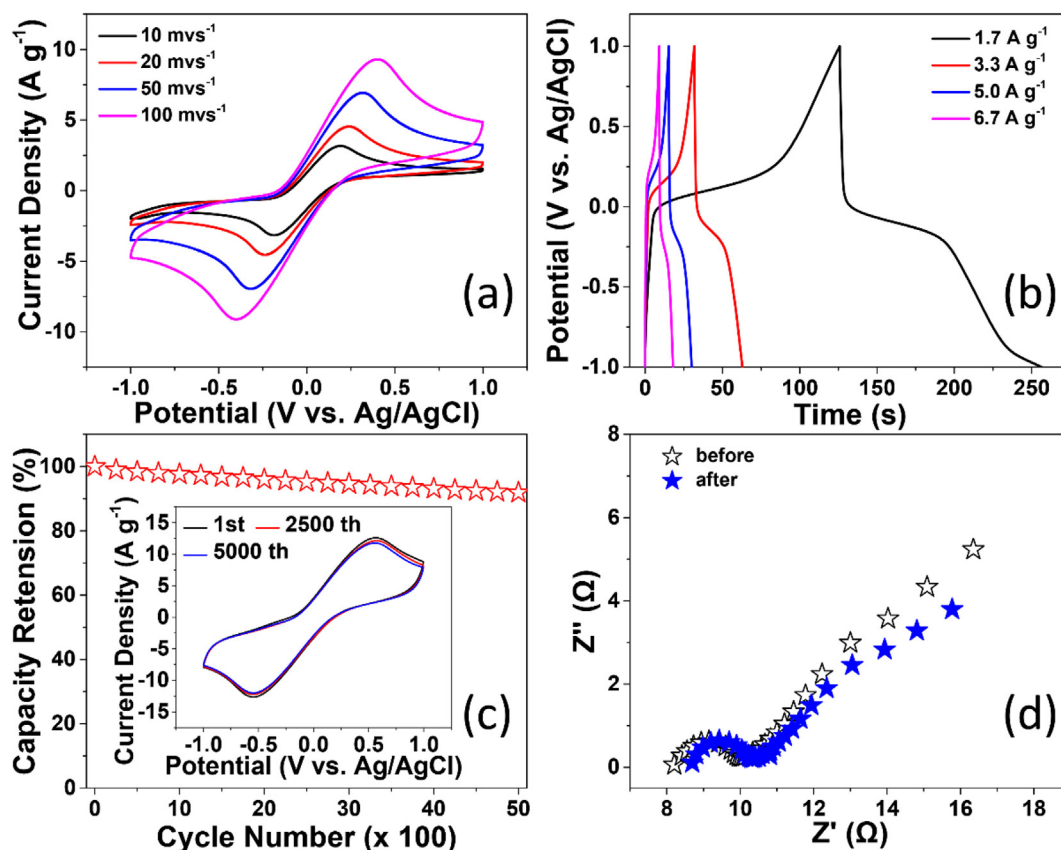


Fig. 5. (a) CVs, (b) GCDs for porous BDD electrode symmetric device recorded in $0.05 \text{ M Fe(CN)}_6^{3-/4-} + 1.0 \text{ M Na}_2\text{SO}_4$ at different scan rates and different current densities. (c) The stability test during 5000-cycle CV scanning, the insets show the 1st, 2500th, and 5000th cycle. (d) Nyquist plot for the porous BDD symmetric device before and after 5000-cycle CV scanning.

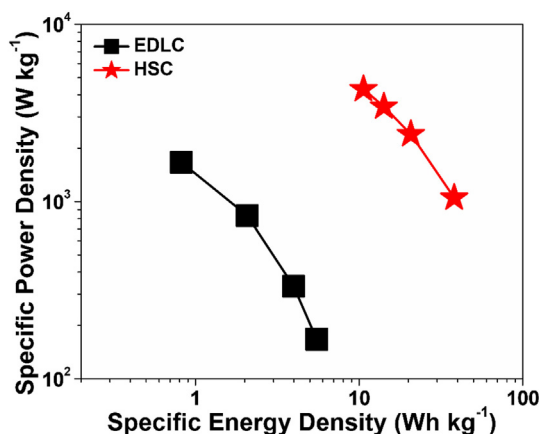


Fig. 6. Ragone plots of the porous BDD symmetric device in $1.0 \text{ M Na}_2\text{SO}_4$ and $0.05 \text{ M Fe(CN)}_6^{3-/4-} + 1.0 \text{ M Na}_2\text{SO}_4$ (mass normalization by porous BDD mass only).

applied, the performance can be hardly compared directly. It is worthy to note that the specific energy/power densities may be overestimated due to only the porous BDD mass is normalized. However, when the redox couples were added to form an HSC, its electrochemical performance is significantly improved compared with that of the EDLC. Thus, it's a new solution to construct high performance and stability HSCs combining porous BDD with water-soluble redox electrolyte. In summary, both the EDLC and HSC devices fabricated from porous BDD electrodes exhibit high power densities and high energy densities, especially when redox couples are introduced as the electrolyte, the HSCs deliver a high specific energy density comparable to some microporous carbon

electrode materials. Together with the long-term capacity retention as well as the large capacity, HSCs are successfully constructed, which can fill the gap between supercapacitors and batteries.

4. Conclusion

A porous BDD film was employed as the capacitor electrode to construct high-performance supercapacitors. The porous BDD film was directly fabricated on a silicon substrate by MPCVD via regrowth of diamond nanoparticles without a template nor a binder. The porous BDD capacitor delivers a high electrical double layer capacitance with excellent stability. To construct HSCs with battery-like performance, a soluble redox couple of $\text{Fe(CN)}_6^{3-/4-}$ was added into the Na_2SO_4 solution. The HSCs feature a large capacity, well stability, high specific energy, and specific power densities, which are comparable to those of reported HSCs. The outstanding electrochemical performance is attributed to the largely enhanced specific areas, the high potential window, the fast charge transferring, the interconnected porous structure, the shorten ion path, as well as the inert nature of the sp^3 bonding structure of diamond. Thus, such a novel strategy to construct HSCs by combining the porous BDD electrode with the soluble redox species will provide a new and valid solution to plug the gap between batteries and supercapacitors.

CRediT authorship contribution statement

Jian Wang: Conceptualization, Methodology, Investigation, Writing - original draft, Writing - review & editing. **Zhibing He:** Methodology, Writing - review & editing. **Xiulan Tan:** Methodology, Investigation. **Tao Wang:** Methodology, Validation. **Xiaoshan He:** Funding acquisition. **Ling Zhang:** Funding acquisition, Writing - review

& editing. **Jinglin Huang**: Formal analysis, Writing - review & editing. **Guo Chen**: Resources. **Kai Du**: Conceptualization, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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